

2025
1st year
cairo dentistry

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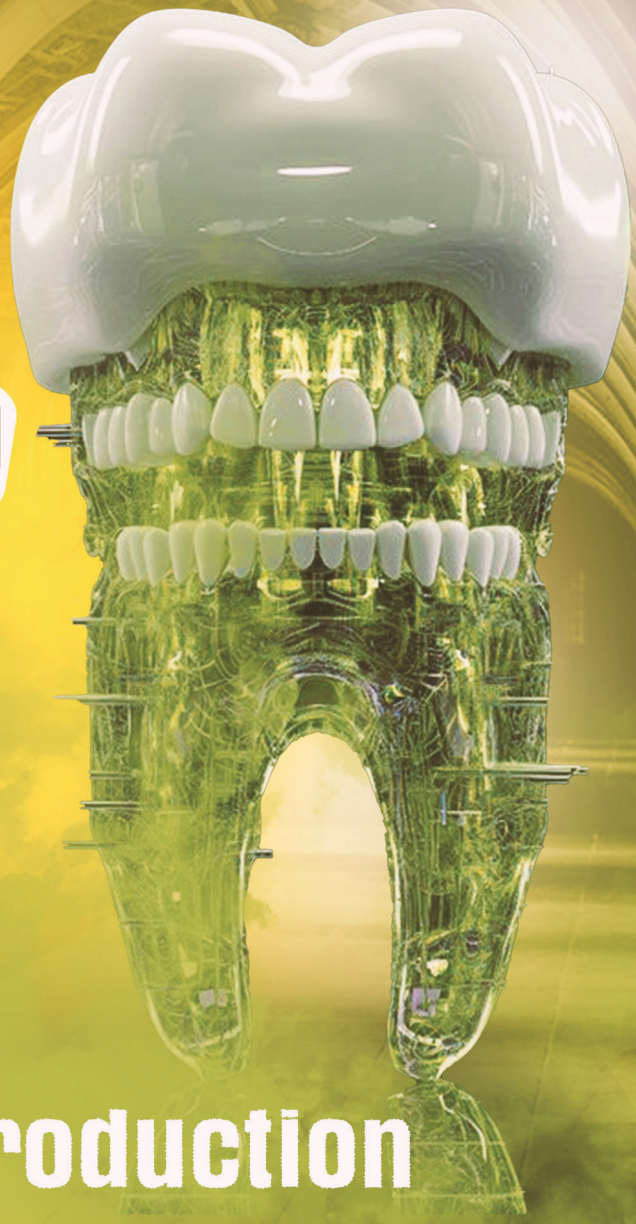
1st term

Pharmaline center

 **01011634691**

1 | **introduction**

Dental Anatomy

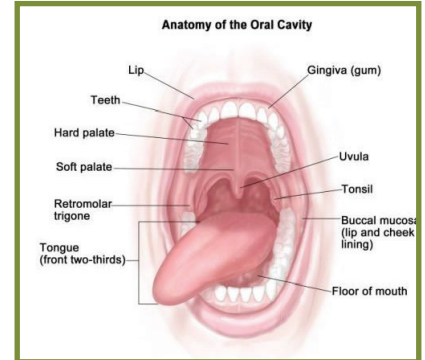


introduction

Dental anatomy includes :

Surface anatomy of teeth ; externally and internally, their management in the dental arches , their relationship to each other and to the skull bone.

Upon looking into the oral cavity you can see :



1 Lips :

- there are 2 lips upper and lower
- the 2 lips meet at the corner of the mouth which is called commissure.
 - a - **Naso labial groove** : run **diagonally** downward on each sides of the nostril toward the corner of the lib.
 - b- **labiomental groove** : run **horizontally** under the lower lip and emphasize the chin



2 Gingiva :

- It is the part of the masticatory tissue that surrounds the cervical part of teeth . it is firmly attached to the teeth and the surrounding bone.



3 Cheek :

- The lining of the inside of the cheeks is shiny with a white line called linea alba buccalis running posteriorly on each side at the level where upper and lower teeth come together .
- The **parotid papilla** : it is a round elevation of tissue observed on the cheek between first and second molar at just above the occlusal plane .
it covers the opening of the parotid duct (**stensen's duct**).



4 Tongue :

- It's the broad flat organ composed of muscle fibers and glands . it rests in the floor of the mouth and manipulate food for mastication. It is the principle organ of taste .



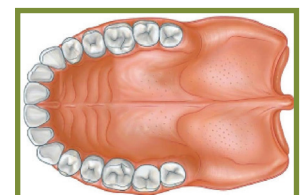
5 Floor of the mouth :

- It is shiny and some large blood vessels seen near the surface.
- Beneath the tongue there are two bilateral bulges caused by the presence of large salivary glands.



6 Roof of the mouth :

- its anterior part is called the **hard palate** . the end of the hard palate is opposite to third molar .
- The soft palate is sometimes reddish than the hard palate . the junction between hard and soft palate is called the vibrating line .
- uvula is a small flesh structure hanging from the center of soft palate .



7 Teeth:

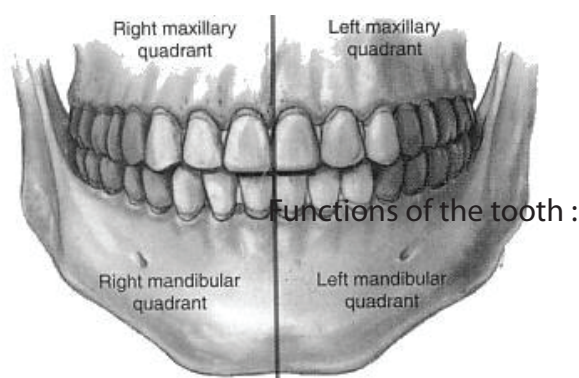
- Human teeth are arranged in the oral cavity in two arches called dental arches one upper and one lower .
- The upper arch is attached to the maxilla . the teeth of upper arch is called upper or maxillary teeth .
- The lower dental arch is attached to a movable bone called the mandible which joins the skull at the temporo- mandibular joint . the teeth of lower dental arch is called the mandibular teeth .
- Each dental arch can be divided into nearly equal halves by an imaginary vertical line called the midline . the right and left halves in each arch are called quadrant . there are four quadrants in the oral cavity .

A maxillary (upper) right quadrant

B maxillary (upper) left quadrant

C mandibular (lower) right quadrant

D mandibular (lower) left quadrant



In the oral cavity there are four types of teeth :

1 Incisors :

they are 2 front teeth in each quadrant
the central incisor is the first tooth next to the midline .
the lateral incisor is the second .

**2 Canines (cuspid) (corner stones):**

There is one canine in each quadrant . it is the third tooth from the midline in each quadrant .

3 Premolars (bicuspid) :

There are 2 premolars in each quadrant , they are the fourth and fifth teeth from midline .
The fourth tooth is the first premolar , the fifth tooth is the second premolar .

4 Molars :

There are 3 molars in each quadrant , they are the sixth , seventh ,and eighth from midline . the sixth is the first molar , the seventh is the second molar , the eighth is the third molar .

Incisors and canine are considered anterior teeth , because they are near the midline .

Premolars and molars are considered posterior teeth , because they are away from the midline .

Periods of dentition

There are 2 types of dentition which are primary and permanent dentitions and 3 phases of dentition which are :

	Primary dentition	Mixed dentition (Ugly duckling stage)	Permanent dentition Secondary dentition Adult dentition
Time	<u>From 6 months to 6 years</u>	<u>From 6 years to 12 years</u>	<u>From 12 years to throughout life</u>
Teeth at this age	- Deciduous teeth only which shed or exfoliate . - The eruption of primary dentation begins at 6 months (for lower incisors) and ends at 2 years (for the upper second deciduous molar)	- Deciduous teeth still exist And permanent teeth erupt - Before eruption of permanent canines , there would be unesthetic appearance , as midline diastema between the 2 incisors . - After the eruption of canines . the diastema closes and so the ugly duckling stage is self-correcting .	All deciduous teeth disappears And there are only permanent teeth.
Definitions	Shedding : it is a physiological loss of deciduous teeth to be replaced by permanent Eruption : it is the appearance of deciduous or permanent teeth in the oral cavity .		

★ The deciduous teeth is **20 in number** , **10 in maxilla** and **10 in mandible** , so 5 in each quadrant as follows :

- 1 Two incisors (A and B) 2 one canine (C) 3 Two molars (D and E)

N.B there are no premolars in deciduous dentition .

The permanent teeth is **32 in number** , **16 in each arch** , **8 in each quadrant** as follows :

- 1 Two incisors (1 & 2) 2 one canine (3)
 3 two premolars (4 & 5) 4 3 molars (6 & 7 & 8)

Deciduous teeth is succeeded by permanent teeth as follows :

(A) Sheds (1) erupts , (B) sheds (2) erupts , (C) sheds (3) erupts , (D) sheds (4) erupts , (E) sheds (5) erupts .

So deciduous molars is succeeded by permanent premolars.

Incisors , canines and premolars are called succdesnous teeth because they are preceded by deciduous teeth while All permanent molars are not preceded with deciduous teeth so they are named as **non succedenous teeth.**

Total number of permanent teeth exceed that of deciduous by 12 (3 molars in each quadrant) .

Dental formula of	Deciduous	I 2/2	C 1/1	Molars 2/2
	Permanent	I 2/2	C 1/1	Premolars 2/2 Molars 3/3

Macro anatomy of the tooth

Each tooth consists of crown , root and a neck (cervix)

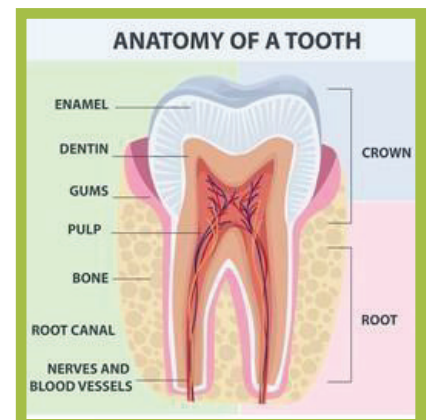
1 Crown:

Anatomical crown : it's the part of the tooth covered with **enamel**.
It remains constant through-out life except if attrition happens .
Attrition is the physiologic loss of hard dental structure by mastication .

Clinical crown : it is the part we can see in the oral cavity above the gingiva .

It may be shorter to the anatomical crown in case of **newly erupted** tooth ,or in case of **gingival swelling** .

It may be longer than the anatomical crown in case of **gingival recession**.



2 Root:

Anatomical root : it is the part of tooth which is covered with **cementum** .

Clinical root : it is the part of the tooth which we **can't** see because it is covered by the gingiva .

N.B. in cases of gingival recession the clinical crown will be **larger than** the **anatomical crown** and the anatomical root is larger than the clinical root .

In case of gingival swelling the **anatomical crown** is larger than the clinical crown and the clinical root is larger than the anatomical root .

- Teeth may have single root or multiple roots . in multiple rooted teeth may be two roots with bifurcation or 3 rooted teeth with trifurcation .

Root Trunk : is the undivided portion of the root .

3 Neck (Cervix)

It is the junction between the anatomical crown and the anatomical root . it is called **the cervical line or the cement-enamel junction** .

Micro anatomy of the tooth

1 Enamel:

Is the **hardest** and the **most mineralized** tissue inside the body . it is also non vital tissue .
It covers the crown to protect it during mastication .

2 Dentin:

It is the **main bulk of the tooth** , it is found in the crown under enamel and also in the root .
It is vital structure so it needs to be covered with enamel .

3 Cementum:

It is the yellowish tissue that covers the anatomical root
, it provides **medium for the periodontal ligament attachment** .

4 Pulp:

It is the **only soft tissue** inside the tooth , it is found in the central part of the tooth surrounded with dentin . It provides the tooth with blood vessels and nerves .

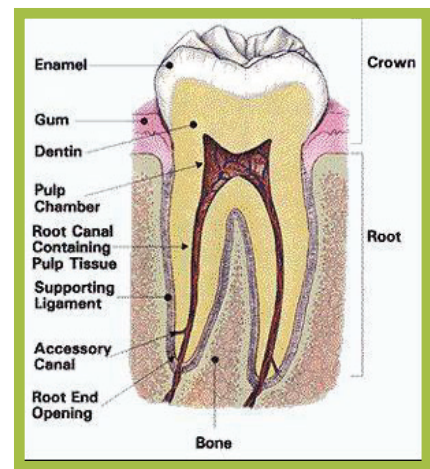
It is formed of 2 parts :

A) Coronal pulp (in pulp chamber) : it is the part of the pulp in the crown , its outline is the same as the external outline of the tooth . it has pulp horns . it is also prominent (large) in young ages .

B) Radicular pulp (in root canals) : it is the part of the pulp inside the roots and ends in the apex of the root by apical foramen .

It may has accessory root canals which is found laterally to the main root canals .

Some roots may have more than one canal which ends in the same apical foramen or separate foramina .



Periodontium : it's the supporting and investing structure of the tooth
It consists of 2 hard tissues: 1) alveolar bone 2) cementum
And 2 soft tissues: 1) gingiva 2) periodontal ligaments

Alveolar bone : it is the part of the bone which surrounds the tooth .

Gingiva : it is the pink tissue surrounds the tooth .

Periodontal ligaments : it is the ligaments between the bone and the root (cementum) .

Junctions of the tooth :

- 1) cement enamel junction (see cervical line)
- 2) dentino enamel junction : junction between the dentin and the enamel , in crown .
- 3) dentino cemental junction : junction between dentin and cementum, in root .



Functions of the tooth :

1) **Mastication** : the most important function .

Incisors : for cutting and incising the food . they are chisel in shape
Canines : for piercing and tearing of food . it is wedge shape

Premolars : the first premolar is for tearing , the second premolar for grinding .

Molars : for grinding of food , because they have wide surface with many elevations (cusps) .

2) **Appearance** : well aligned teeth gives good appearance , the facial appearance changes if anterior teeth extracted

3) **Speech** : for clear pronunciation and good production of sounds .

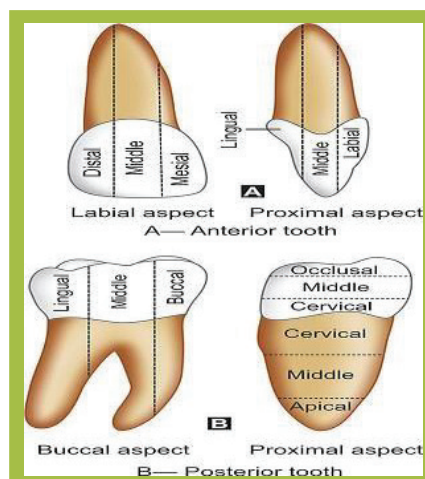
4) **Growth of jaws** : teeth have a role in growth of jaws in some periods . if a child eats on one side , the facial growth will be unilateral .

Surfaces of teeth : each tooth has five surfaces : 1) facial (labial for anteriors and buccal for posterior) , 2) lingual (for lower and upper teeth and palatal for upper only) , Proximals : 3) Mesial which is near the midline & 4) distal which is away from the midline) , and 5) masticatory surface : incisal (for anteriors) or (occlusal for posteriors) .

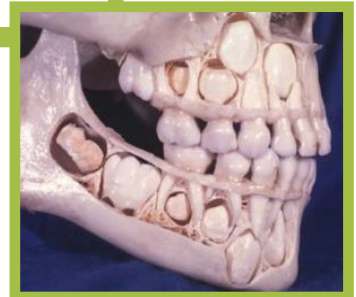
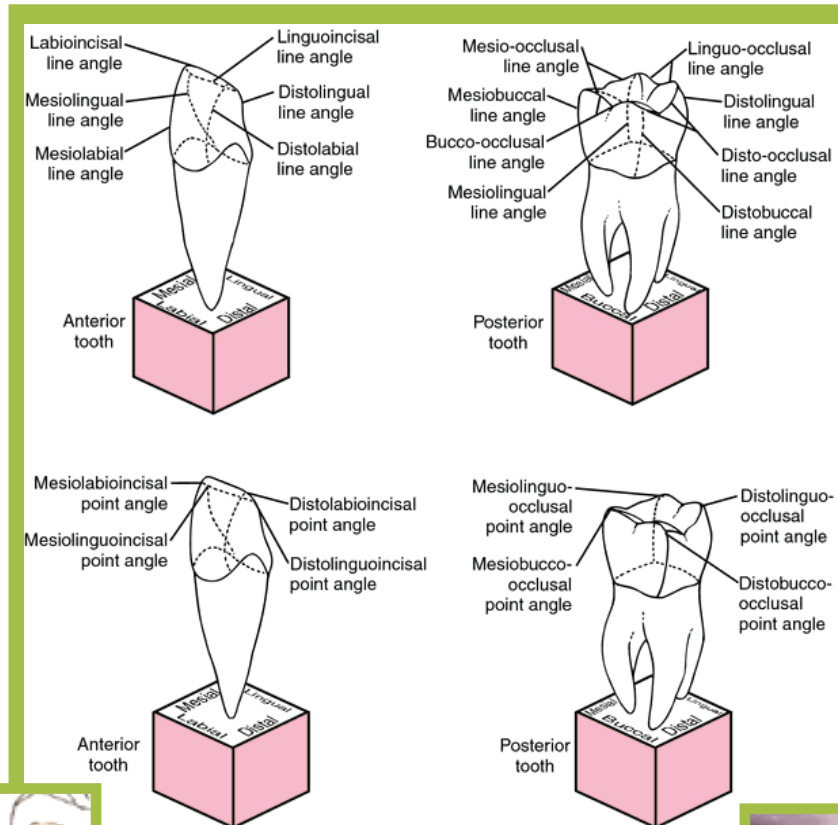
N.B Incisal for biting .. occlusal for chewing.

Palatal surface is so named because it faces the hard palate . while lingual because it faces the tongue .

Division into thirds :



Line angles and point angles



	anterior teeth	posterior teeth
line angle	6	8
point angle	4	4

	Definition
Socket or Alveolus	It is the bony space in the jaw containing the roots of the erupted tooth.
Crypt	It is the bony space in the jaw containing the developing unerupted tooth.



Tooth Numbering

1) Palmer Notation System:

For permanent		For Deciduous	
Upper right	Upper left	upper right	upper left
8 7 6 5 4 3 2 1	1 2 3 4 5 6 7 8	E , D , C , B , A	A , B , C , D , E
8 7 6 5 4 3 2 1	1 2 3 4 5 6 7 8	E , D , C , B , A	A , B , C , D , E
Lower right	Lower left	Lower right	Lower left

2) International numbering system (2 digits system)

For permanent	
Upper right	Upper left
18 17 16 15 14 13 12 11	21 22 23 24 25 26 27 28
48 47 46 45 44 43 42 41	31 32 33 34 35 36 37 38
Lower right	Lower left

For Deciduous	
Upper right	Upper left
55 54 53 52 51	61 62 63 64 65
85 84 83 82 81	71 72 73 74 75
Lower right	Lower left

3 , D

what is tooth number 45 and what system



2) The Universal numbering system (American numbering system) : # or #d

For permanent

Upper right								Upper left							
1 2 3 4 5 6 7 8								9 10 11 12 13 14 15 16							
32 31 30 29 28 27 26 25								24 23 22 21 20 19 18 17							
Lower right								Lower left							

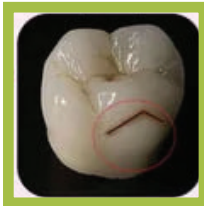
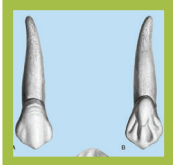
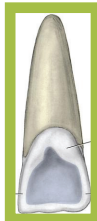
For Deciduous

Upper right					Upper left					Upper right					Upper left				
1 2 3 4 5					6 7 8 9 10					A B C D E					F G H I J				
20 19 18 17 16					15 14 13 12 11					T S R Q P					O N M L K				
Lower right					Lower left					Lower right					Lower left				

Test yourself

- What is the tooth # 15 and what system What is the tooth 15 and what system ?
- What is the tooth #d

Anatomical landmarks

Elevation	Lobe		Tubercle	Ridges
Definition	- It is one of the primary center of calcification formed during the crown development - All teeth have 4 lobes as : - Anteriors : 3 labial + 1 lingual - All premolars : 3 buccal + 1 lingual except lower 5 : 3 buccal + 2 lingual = 5 - upper and lower molar : 2 buccal + 2 lingual Except lower 6 : 3 buccal + 2 lingual = 5 - only upper 8 & upper lateral (peg shape) : have 3 lobes only .		Small rounded Elevation. Formed of Enamel only but in excessive form . 	■ A) Ridges on facial surface : labial ridge : on middle third of labial aspect of upper and lower canines Buccal Ridge : on middle third of buccal aspect of all premolars Cervical ridge : on cervical third of all teeth , but it is well developed on upper D followed by E . ■ B) Ridges on lingual surface : Incisal Ridge : found in lingual aspect Of newly erupted incisors at the incisal margin , the ridge is transformed into edge by attrition . Lingual ridge : found in lingual surface of upper canine and is well developed (from cusp tip to cingulum) and in lower canine and is less developed (confined in incisal 1/3 only . cusp ridge : they are called cusp slopes or Cusp arms , usually distal cusp slope > mesial slope. Except in upper C and upper 4 (buccal cusp) Marginal ridge : found lingually in anterior Teeth and occlusally in posterior teeth .
Found in	Mature forms of lobes includes : Mamelons , cusp , and cingulum		- Upper 6 on Mesio-lingual Cusp - Upper E - sometimes Over the Cingulum .	
	1-Mamelons	2-Cusp	3-Cingulum	
Definition	3 rounded Small Elevations Of enamel. The pulp Chamber of Maxillary Incisors Have 3 pulp Horns Correspond To the 3 mamelons	It is pyrimidal elevation formed of enamel + dentin + pulp	It's a rounded elevation known as Lingual Cervical Ridge .	Ridges on occlusal surface : triangular ridge : it is found in posterior Teeth descending from the cusp tip to the center . each cusp has one triangular ridge , except mesiolingual cusp of upper 6 has 2 triangular ridges transverse ridge : it is the union of 2 triangular ridges transversely of lower premolars only . oblique ridge : it's union of 2 triangular Ridges oblique , found in upper 6 & upper E Sometimes in upper 7 & 8 & D
Found in	In Newly erupted permanent incisors . in incisal 1/3 	In canine Incisally And in all Posteriors Occlusaly 	Only In anterior teeth Lingually In cervical 1/3 	

Crown depressions

Circumscribed

Linear

Fossa

pit

Definition

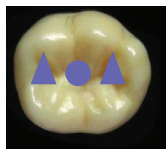
Circumscribed depression

True pit

It is a small pinpoint depression .

Faulty pit

It is incomplete formation of enamel .

Found inIn all teeth :
Lingually in anteriors ,
It is named as lingual FossaOcclusally in posterior
Teeth, mesial and
Distal Triangular fossa

- At the end of central developmental grooves
- At the bottom of central fossa (central pit)
- At the bottom of mesial and distal triangular fossae (mesial and distal pits)

- buccal Developmental groove of lower molar .
- palatal Developmental groove of upper of upper molars

1) developmental groove :

It's a shallow linear depression , due to Complete union of lobes It's named as the surface it founds , so There is central developmental groove Buccal and lingual developmental groove

2) Fissure :

It is a linear depression found in the bottom of central developmental groove it's result from incomplete union between lobes . is's a fault in enamel it may result in caries.

3) Supplemental grooves :

It is found in occlusal surface of upper 8 and lower 8 followed by 7 . giving them wrinkled appearance. It does not represent union between lobes .

4) Sulcus

It's a valley between the inclined planes Of triangular ridges on occlusal surface of posterior teeth . it inclines from the cusp tip to the developmental groove



Life history of the tooth

1|Soft growth :

The deciduous teeth start their development inside the jaw before birth , from 6 w.i.u the oral epithelium began to form the germ of 10 deciduous teeth in each jaw .

Most of permanent teeth start their development inside the jaw before birth .

The tooth germ is formed of soft tissue and does not appear in x-ray films .

2|Calcification :

Calcification of hard dental tissues as enamel , dentin and cementum occurred by specialized cells .

The mineralized salts , mostly calcium , from blood will deposit in the soft tissue in form of small spherical bodies , then they will enlarge to form primary lobes which fuses together to form calcified dental tissues .



3 Eruption :

- The eruption of mandibular teeth precedes the maxillary teeth .
- The teeth in both sides of the jaw erupt at the same time .
- Teeth erupt in girls earlier than boys .
- The teeth will erupt in the oral cavity when the crown is completely formed as well as third of the root .

4 Attrition :

It's a physiological loss of hard dental structure due to the continuous friction of teeth together during mastication .

The attrition is clearly seen in incisal and occlusal surfaces also at the contact point .

The contact point between teeth is lost and changes to contact area due to attrition

4 Shedding of deciduous teeth :

It's a physiological loss of deciduous teeth and replacing it with permanent ones .

Before shedding of deciduous teeth by 3 years , their roots began resorption apically till reach their whole length then their crowns are exfoliated .

Loss of permanent teeth or deciduous teeth before their shedding time is considered a pathological loss of teeth .

THANK YOU ...